Using u-Substitution in Indefinite and Definite Integrals

AP Calculus AB / BC   Integration Techniques

Recognising an inner function and its derivative, adjusting by a constant factor, converting the limits of a definite integral to the new variable, and evaluating an improper integral as a limit

The one idea behind every problem. u-substitution is the chain rule read backwards. If an integrand contains an inner function *and* a constant multiple of its derivative, naming that inner function u turns the whole integral into one you already know.

The substitution box — fill one in before integrating

$u =$	$du = \qquad dx$
new lower limit	new upper limit

Indefinite integrals leave the limit row blank and back-substitute at the end. Definite integrals fill it in and never return to x.

Reference layout. No values are filled in — use the integrand given in each problem.

Directions.

- Write the substitution box first, every time. Most errors here are chosen, not computed.
- Include $+C$ on every indefinite integral.
- Give exact answers. Write your final answer on the answer line.

1. Evaluate $\displaystyle\int 2x\left(x^2+5\right)^3\,dx$.

(a) Fill in the substitution box: what is u, and what is du?

(b) Notice that $2x\,dx$ is exactly du — no adjusting factor is needed. Rewrite the integral entirely in u.

(c) Integrate and back-substitute.

Answer: ________________

2. Evaluate $\displaystyle\int x\cos(x^2)\,dx$.

(a) Take $u = x^2$. Write du, then compare it with the $x\,dx$ that actually appears in the integrand.

(b) What constant must multiply the integral to make up the difference? Explain in one sentence why moving a constant across the integral sign is legal but moving an x is not.

(c) Finish the integration and back-substitute.

Answer: __________________

3. Evaluate $\displaystyle\int \frac{(\ln x)^2}{x}\,dx$.

(a) The integrand contains a function and its own derivative. Which factor is which?

(b) Fill in the substitution box and rewrite the integral in u.

(c) Integrate and back-substitute.

Answer: __________________

4. Evaluate $\int_0^2 x\, e^{x^2}\, dx$ exactly.

(a) With $u = x^2$, fill in *all four* cells of the substitution box, including the two new limits.

(b) Rewrite the definite integral in u with its new limits. You should not need to return to x at any point.

(c) Evaluate. Leave your answer in terms of e.

Answer: ____________________

5. Evaluate $\int \sin^3 x\, \cos x\, dx$.

(a) Two substitutions look plausible here: $u = \sin x$ and $u = \cos x$. Try writing du for each.

(b) Only one of them lets you replace *every* x in the integrand. Say which, and why the other one strands a leftover factor.

(c) Carry out the substitution that works, integrate, and back-substitute.

Answer: ____________________

6. Evaluate $\displaystyle\int_1^3 \frac{2x}{x^2+1}\,dx$ exactly.

(a) Fill in all four cells of the substitution box.

(b) Evaluate the integral in u. Your first answer will be a difference of two logarithms.

(c) Simplify it to a single logarithm.

Answer: __________________

7. Evaluate $\displaystyle\int x^2\sqrt{x^3+1}\,dx$.

(a) Fill in the substitution box, then state the constant factor the integral must carry.

(b) Rewrite $\sqrt{u}$ as a power before integrating.

(c) Integrate, simplify the two fractions into one coefficient, and back-substitute.

Answer: __________________

8. Challenge. Consider the improper integral $\int_0^\infty x\, e^{-x^2}\, dx$.

(a) An infinite limit is not a number, so rewrite the integral as a limit of a proper definite integral on $[0, b]$.

(b) Use $u = -x^2$ or $u = x^2$ to find the antiderivative, evaluate it on $[0, b]$, and take the limit as $b \to \infty$. What is the value of that limit?

(c) State whether the integral converges or diverges, and give its value. Write your part (b) limit and your part (c) value on the answer line.

Answer: ____________________